

الاسم والشهرة :-----

رقم المرشح :-----

الإمتحانات الرسمية لشهادة الاجازة الفنية
لدورة عام 2021 الأولى

الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

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الإختصاص والرمز : المعلوماتية : فرع البرمجيات (6807)

المادة : حماية الشبكات ونظام هيكليّة المستخدم / الموزع

المستندات المسموح بها : لا شيء

المعدل : (8)

المدة : ساعة ونصف

الإجابات على "كراس الإجابة"

I- Choose the correct answer: (1 pt/correct answer)

1- An RPC (remote procedure call) is initiated by the: a- Server b- Client c- Middleware d- a and b	2- Which of the following is an example of a client-server model? a- FTP b- DNS c- TELNET d- All of the above
3- The use of RPCs is less efficient than the direct use of sockets: a- True b- False	4- Sockets use a well-known port to contact a remote server, but RPC clients only know the remote RPC program ID and procedure number: a- True b- False
5- A client device designed so that most processing and data storage occurs on the server side describes: a- Thin Client b- Fat Client c- Passive Client d- Active Client	6- A socket is a point of communication between the client process and the server process, it is the combination of: a- An IP address and a port number b- An IP address and a MAC address c- A MAC address and a port number d- A MAC address, a port number and an IP address.
7- The process of transforming plain text into unreadable text. a- Decryption b- Encryption c- Network Security d- Information Hiding	8- An encryption technique with 2 keys is : a- Monoalphabetic Cipher b- Steganography c- Symmetrical cryptography d- Asymmetrical cryptography
9- In Asymmetrical cryptography, a key that decrypts the message is a : a- Public key b- Unique key c- Private key d- Security key	10- In a corporate environment, what is the best measure to mitigate social engineering attacks? a- Install next generation firewalls b- Make sure that all end user devices are properly patched and updated c- Regularly train employees and make sure that they are aware of the security policy and the acceptable use policy d- Perform auditing, penetration testing, and vulnerability assessments

II- 1- For a large-scale company, what are the main advantages of a client / server system? (2 pts)

2- In the client / server architecture, if the server is the single point of failure in the network. (1 pt)

Suggest a solution for this problem. Give examples.

III- Give the characteristics of the model at N levels (Components) (3 pts)

IV- Being an e-commerce business. This company sells items on its site. (4 pts)

A customer who chooses items specifies his address for delivery and pays using his credit card.

1- A customer who accesses the website must specify his identity and address.

A hacker can change this request to put his name and address.

What security property is under attack? How do we solve this problem?

2- A customer who has to pay the amount, sends his credit card information.

A hacker can steal this information. What security property is under attack? How do we solve this problem?

3- A server may deny that it has received a purchase request.

What security property is under attack? How do we solve this problem?

4- A hacker can send a series of control messages to this site to stop his service.

What security property is under attack? How do we solve this problem?